

Final Exam R4

1. If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable at $x \in \mathbb{R}^n$, prove that $Df(x) = \nabla f(x)$. **1p**
2. Consider the curve defined by the equation $f(x, y) = -8$, where $f(x, y) = 3x^2 - 12x + 2y^2 - 4y$.
 - (a) Find the direction (unit vector) in which f increases the most at the point $(1, 0)$. **0.5p**
 - (b) Write a gradient descent method to find the minimum of f , starting from $(1, 0)$. **0.5p**
 - (c) Find the equation of the tangent line to the curve at the point $(1, 0)$. **1p**
3. Consider the function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, $f(x, y, z) = x^3 + y^3 + z^3 - 3x^2 - 3y^2 - 3z^2 + 6xy + 6yz + 6xz$.
 - (a) How many critical points does the function f have? **1p**
 - (b) Classify the critical point $(0, 0, 0)$. **1p**
 - (c) Write the second order Taylor polynomial for f centered around $(0, 0, 0)$. **1p**
4. Let $A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & 1 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, $\alpha > 0$. Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, $f(x) = \|Ax - b\|^2 + \alpha\|x\|^2$.
 - (a) Find the unique $x^* \in \mathbb{R}^3$ that solves the problem: $\min_{x \in \mathbb{R}^3} \|x\|^2$ subject to $Ax = b$. **1p**
 - (b) Find the unique $x_\alpha^* \in \mathbb{R}^3$ that minimizes f over $\mathbb{R}^3$. Show that $\lim_{\alpha \rightarrow 0} x_\alpha^* = x^*$. **1p**
5. Find $\iint_D \ln(1 + x^2 + y^2) dx dy$, where $D = \{(x, y) \in \mathbb{R}^2 \mid y \geq 0, x^2 + y^2 \leq 4\}$. **2p**

Time: 2h. Total number of points: 10p. Please write your full name, your group and your row number R4.